- **Why do we use the value 1.5 in the outlier detection formulas Q1 – (1.5 × IQR) and Q3 + (1.5 × IQR)?**
  Where did this "1.5" come from?

  - The number 1.5 was suggested by a statistician named John Tukey when he created the boxplot method to detect unusual values in data.
  - After testing many datasets, statisticians noticed that 1.5 × IQR works well to catch values that are unusually far from most data but still doesn't wrongly mark normal values as outliers.
  - So, 1.5 became a commonly accepted rule, because it gives a reasonable distance to detect unusual values without marking normal values as outliers.

a. The interquartile range. Compare the two interquartile ranges.

b. Any outliers in either set.

The five number summary for the day and night classes is

	Minimum	Q_1	Median	Q_3	Maximum
Day	32	56	74.5	82.5	99
Night	25.5	78	81	89	98

a. IQR of Day = Q3 - Q1
= 82.5 – 56
= 26.5

IQR of Night = Q3 – Q1
= 89 – 78
= 11

- ✓ **The middle 50% of the day class scores are spread over 26.5 points, while the night class scores are spread over 11 points.**
- ✓ **This shows that the day class has more variation than the night class.**

b. Lesser Outlier Day = Q1 – (1.5*IQR)
= 56 – (1.5*26.5)
= 16.25

Greater Outlier Day = Q3 + (1.5*IQR)
= 82.5 + (1.5*26.5)
= 122.25

- ✓ **The day class has no outliers because the minimum value (32) and maximum value (99) fall within the calculated limits of 16.25 and 122.25.**

Lesser Outlier Night = Q1 – (1.5*IQR)
= 78 – (1.5*11)
= 61.5

Greater Outlier Night = Q3 + (1.5*IQR)
= 89 + (1.5*11)
= 105.5

- ✓ **The night class has a lower outlier because the minimum value (25.5) is less than the lower boundary (61.5).**
- ✓ **There is no greater outlier because the maximum value (98) is within the upper boundary (105.5).**